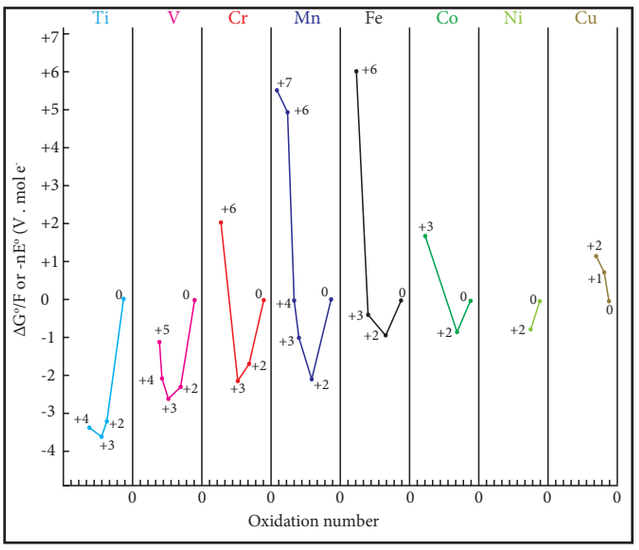
having a stable +1 oxidation state. It is prone to disproportionate to the +2 and 0 oxidation states.

**Evaluate yourself:**

Why iron is more stable in +3 oxidation state than in +2 and the reverse is true for Manganese?



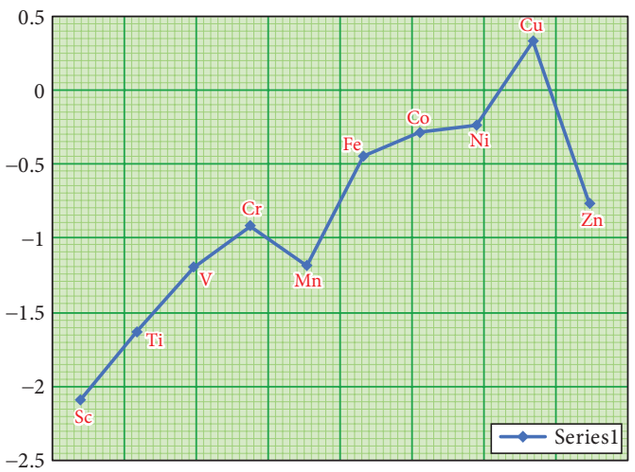

### 4.3.5 Standard electrode potentials of transition metals

Redox reactions involve transfer of electrons from one reactant to another. Such reactions are always coupled, which means that when one substance is oxidised, another must be reduced. The substance which is oxidised is a reducing agent and the one which is reduced is an oxidizing agent. The oxidizing and reducing power of an element is measured in terms of the standard electrode potentials.

Standard electrode potential is the value of the standard emf of a cell in which molecular hydrogen under standard pressure ( 1atm) and temperature (273K) is oxidised to solvated protons at the electrode.

If the standard electrode potential ($E^0$), of a metal is large and negative, the metal is a powerful reducing agent, because it loses electrons easily. Standard electrode potentials (reduction potential) of few first transition metals are given in the following table.

| Reaction | Standard reduction potential ( V ) |
|---|---|
| $Ti^{2+} + 2e^- \longrightarrow Ti$ | −1.63 |
| $V^{2+} + 2e^- \longrightarrow V$ | −1.19 |
| $Cr^{2+} + 2e^- \longrightarrow Cr$ | −0.91 |
| $Mn^{2+} + 2e^- \longrightarrow Mn$ | −1.18 |
| $Fe^{2+} + 2e^- \longrightarrow Fe$ | −0.44 |
| $Co^{2+} + 2e^- \longrightarrow Co$ | −0.28 |
| $Ni^{2+} + 2e^- \longrightarrow Ni$ | −0.23 |
| $Cu^{2+} + 2e^- \longrightarrow Cu$ | +0.34 |
| $Zn^{2+} + 2e^- \longrightarrow Zn$ | −0.76 |




In 3d series as we move from Ti to Zn, the standard reduction potential $\left(E^0_{M^{2+}/M}\right)$ value is approaching towards less negative value and copper has a positive reduction potential. i.e.,



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